

Diverging Local Bandwidth Suffices for the Singular-Continuous Three-Limit Lower Bound

Candidate strong partial result for arXiv:1908.06721

21 July 2026

Status. Candidate strong partial result, likely valid, awaiting human review. The theorem removes the source's square-root growth rate, but it does not settle the bounded-bandwidth case.

1 Source question

The source is Matthew J. Colbrook, *Computing Spectral Measures and Spectral Types* [1]. Its identifier is arXiv:1908.06721v3. For an integer-valued function f with $f(n) > n$, let P_n be projection onto the first n canonical basis vectors and let

$$\Omega_{f,0} = \{T = T^* : (I - P_{f(n)})TP_n = 0 \text{ for every } n\}.$$

Writing $\Xi_{\text{sc}}^{\mathbb{C}}(T) = \sigma_{\text{sc}}(T)$, the source proves in Theorem 5.1 that

$$\{\Xi_{\text{sc}}^{\mathbb{C}}, \Omega_{f,0}, \Lambda_i\} \notin \Delta_3^G, \quad i = 1, 2,$$

provided $f(n) - n \geq \sqrt{2n} + 1/2$. In the summary on page 6 it asks whether this growth condition can be dropped.

We also consider the computation of spectra as sets in the complex plane. Convergence is measured using the Hausdorff metric in the bounded case and using the Attouch–Wets metric in the unbounded case (i.e. uniform convergence on compact subsets of $\mathbb{C}$). Specifically, we provide the first algorithms computing these quantities and prove in **Theorem 5.1** that:

- The absolutely continuous spectrum $\sigma_{\text{ac}}(T)$ can be computed in two limits but not one limit ($\Delta_3^A \setminus \Delta_2^G$ classification).
- The pure point spectrum $\sigma_{\text{pp}}(T)$ can be computed in two limits but not one limit ($\Delta_3^A \setminus \Delta_2^G$ classification).
- The singular continuous spectrum $\sigma_{\text{sc}}(T)$ can be computed in three limits (Δ_4^A classification). If $f(n) - n \geq \sqrt{2n} + 1/2$, then the computation cannot be done in two limits ($\notin \Delta_3^G$ classification). That is, if the local asymptotic bandwidth is allowed to grow sufficiently rapidly, three limits are needed, and this computational problem is exceedingly difficult. We do not know whether this growth condition on f can be dropped. However, without it, the problem still requires at least two limits ($\notin \Delta_2^G$ classification).

Figure 1: The exact question in the source paper, page 6.

The present packet proves the same exclusion whenever $f(n) - n \rightarrow \infty$. It also identifies why the remaining bounded-bandwidth case cannot be reached by merely improving the enumeration in the source's direct-sum proof.

2 Proof intuition

Colbrook reduces a height-three decision problem to singular-continuous spectrum by taking a direct sum of countably many half-line Schrödinger operators. The displayed square-root rate enters only when that disjoint union of rays is placed into one copy of $\ell^2(\mathbb{N})$ using a triangular enumeration.

If $h(n) = f(n) - n$ tends to infinity, one can do better. Wait until the entire tail has $h \geq k$, and then enumerate the first k rays round-robin. The gap between consecutive vertices of any active ray is exactly k , so every edge crossing a cut at n ends no later than $n + h(n) = f(n)$. Increasing k slowly gives a bijection with all countably many rays and no prescribed rate of divergence.

This mechanism is optimal for the direct-sum route. Across the cut after n , every ray that has already started but is infinite needs a distinct crossing endpoint among the $h(n)$ sites $n+1, \dots, f(n)$. As the number of started rays tends to infinity, necessarily $h(n) \rightarrow \infty$. Thus bounded bandwidth requires a genuinely different spectral encoding.

3 The adaptive enumeration

We first isolate the combinatorial point. A *ray enumeration* is a bijection between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$; for fixed j , the pairs (j, m) , $m = 1, 2, \dots$, are the successive vertices of ray j .

Lemma 1 (Adaptive round-robin enumeration). *Let $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $f(n) > n$, and put $h(n) = f(n) - n$. If $h(n) \rightarrow \infty$, there is a ray enumeration with the following property: if $p < q$ are the positions of two successive vertices of one ray, then*

$$q \leq f(n) \quad (p \leq n < q). \quad (1)$$

Proof. Choose integers

$$1 = c_1 < c_2 < c_3 < \dots$$

recursively so that

- (i) $h(n) \geq k$ for all $n \geq c_k$;
- (ii) $c_{k+1} - c_k$ is a positive multiple of k .

This is possible because $h(n) \rightarrow \infty$: after choosing c_k , move far enough into the tail where $h \geq k+1$ and then move finitely farther to the required congruence class modulo k .

On the phase $c_k \leq n < c_{k+1}$, label the positions periodically

$$1, 2, \dots, k, 1, 2, \dots, k, \dots$$

Condition (ii) makes the phase an integral number of cycles. At phase k the labels $1, \dots, k$ occur, and every fixed label occurs in every later phase; therefore, numbering the successive occurrences of label j by $m = 1, 2, \dots$ gives a bijection $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$.

Within phase k , consecutive occurrences of a fixed label are k positions apart. At the transition from phase k to phase $k+1$, the last occurrence of an old label $j \leq k$ is at

$$c_{k+1} - k + j - 1,$$

and its first occurrence in the next phase is at $c_{k+1} + j - 1$; that gap is again k . New label $k+1$ has no predecessor. Consequently, if an edge $p < q$ starts in phase k , then $q - p \leq k$. For $p \leq n < q$, condition (i) gives $h(n) \geq k$, and hence

$$q - n \leq q - p \leq k \leq h(n).$$

This is exactly (1). □

Proposition 1 (Sharpness for disjoint rays). *Suppose a ray enumeration satisfies (1). Then $f(n) - n \rightarrow \infty$.*

Proof. For a fixed cut n , let A_n be the set of rays having at least one vertex among positions $1, \dots, n$. For each $j \in A_n$, take the last vertex of ray j at or before n and the next vertex of that ray. The latter is a crossing endpoint q_j satisfying

$$n < q_j \leq f(n).$$

The endpoints q_j are distinct because the enumeration is a bijection. Therefore

$$|A_n| \leq f(n) - n.$$

Every one of the countably many rays eventually starts, so for every K one has $|A_n| \geq K$ for all sufficiently large n . Thus $f(n) - n \rightarrow \infty$. $\square$

4 Strengthened lower bound

We use the source notation Λ_1 for matrix-entry evaluations and Λ_2 for the augmented evaluation set that also contains the relevant quadratic matrix data. The SCI classes and the metric on closed subsets of $\mathbb{C}$ are exactly those of [1].

Theorem 1 (Diverging-bandwidth lower bound). *Let $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $f(n) > n$ and*

$$f(n) - n \longrightarrow \infty.$$

Then, for $i = 1, 2$,

$$\{\Xi_{\text{sc}}^{\mathbb{C}}, \Omega_{f,0}, \Lambda_i\} \notin \Delta_3^G.$$

Equivalently, no height-two tower of general algorithms computes the singular-continuous spectrum on $\Omega_{f,0}$.

Proof. We repeat the reduction in the proof of Theorem 5.1 of [1], changing only its enumeration of the direct sum.

Let $a = (a_{k,j})_{k,j \in \mathbb{N}} \in \{0,1\}^{\mathbb{N} \times \mathbb{N}}$ and consider the decision problem

$$\Xi'(a) = 1 \iff \text{some column of } a \text{ contains infinitely many ones.}$$

This problem has general SCI equal to 3; this is the decision problem used in the source proof (with the lower bound cited there from Ben-Artzi–Hansen–Nevanlinna–Seidel [3]).

For every column j , define on $\ell^2(\mathbb{N})$

$$H_j = H_0 + v_j, \quad v_j(m) = \sum_{k=1}^{\infty} a_{k,j} \delta_{m,k!},$$

where H_0 is the free half-line discrete Schrödinger operator. The sparse potential theorem of Krutikov and Remling [2], in precisely the form quoted as Theorem 3.4 in [1], gives

$$\begin{aligned} [0, 4] &\subset \sigma_{\text{sc}}(H_j) && \text{if column } j \text{ has infinitely many ones,} \\ \sigma_{\text{sc}}(H_j) \cap (0, 4) &= \emptyset && \text{if column } j \text{ has only finitely many ones.} \end{aligned}$$

Set $\tilde{H} = \bigoplus_{j=1}^{\infty} H_j$. Use Lemma 1 to identify its canonical basis with that of $\ell^2(\mathbb{N})$. The only off-diagonal entries of every H_j join successive vertices of its ray. If such an edge has positions $p < q$, then for every cut $p \leq n < q$, Lemma 1 gives $q \leq f(n)$. Hence no column among the first n has a nonzero entry below row $f(n)$, and therefore

$$(I - P_{f(n)})\tilde{H}P_n = 0 \quad \text{for every } n.$$

Thus $\tilde{H} \in \Omega_{f,0}$. Its entries are uniformly bounded, so it is a bounded self-adjoint operator.

The direct-sum spectral decomposition now yields

$$\Xi'(a) = 1 \implies [0, 4] \subset \sigma_{\text{sc}}(\tilde{H}), \quad \Xi'(a) = 0 \implies \sigma_{\text{sc}}(\tilde{H}) \cap (0, 4) = \emptyset.$$

Every queried matrix entry of $\tilde{H}$ depends on at most one entry of a , after applying the fixed enumeration and its inverse. The augmented Λ_2 data are also finite-information functions of a , because every column of $\tilde{H}$ has at most three nonzero entries. Thus the reduction works for both Λ_1 and Λ_2 .

If a height-two general tower computed $\sigma_{\text{sc}}(\tilde{H})$, applying the continuous postprocessing

$$F \longmapsto 1 - \min\{1, \text{dist}(3, F)\}$$

and then the discrete-output conversion in the source proof of Theorem 5.1 would give a height-two general tower for Ξ' . This contradicts $\text{SCI}_G(\Xi') = 3$. The claimed exclusion follows. $\square$

5 Verification and limitations

Internal checks. The script `code/check_round_robin.py` constructs the phases for the slowly diverging sample

$$h(n) = \lfloor \log_2(n+1) \rfloor + 1$$

and checks every cut inequality in a finite prefix. This is only a sanity check; Lemma 1 is the proof. The most important points for human verification are (a) the phase-transition gap is k , not $k+1$, and (b) the cut inequality is exactly the matrix condition defining $\Omega_{f,0}$.

Scope. The theorem removes every quantitative growth rate but retains $f(n) - n \rightarrow \infty$. Proposition 1 shows that the latter cannot be removed from this direct-sum/ray-enumeration proof. The original question remains open for bounded or non-diverging local bandwidth, notably $f(n) = n+1$.

Novelty check. The run’s lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:1908.06721, the title, singular-continuous spectrum, SCI, and bandwidth terminology; no duplicate was found. A bounded web/arXiv search on 21 July 2026 used the exact source sentence and the phrases “singular continuous spectrum”, “Delta 3”, “SCI”, “bandwidth”, and “Colbrook”. It found the source and later numerical spectral-measure papers but no separate settlement of this hypothesis. The novelty confidence is moderate rather than definitive because this was not an exhaustive citation review.

Recommendation. Promote as a strong partial result if a reviewer confirms Lemma 1 and the finite-information simulation for Λ_2 . Do not describe it as a full answer to the source question.

References

- [1] M. J. Colbrook, *Computing Spectral Measures and Spectral Types*, Communications in Mathematical Physics 384 (2021), 433–501; arXiv:1908.06721v3.
- [2] D. Krutikov and C. Remling, *Schrödinger operators with sparse potentials: asymptotics of the Fourier transform of the spectral measure*, Communications in Mathematical Physics 223 (2001), 509–532.
- [3] J. Ben-Artzi, M. J. Colbrook, A. C. Hansen, O. Nevanlinna, and M. Seidel, *Computing Spectra—On the Solvability Complexity Index hierarchy and towers of algorithms*, arXiv:1508.03280v5; see also the exact decision-problem citation in [\[1\]](#).